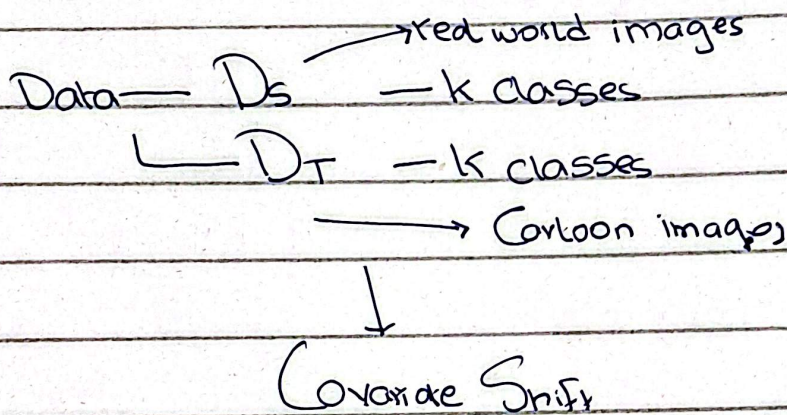
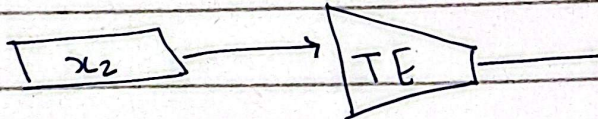
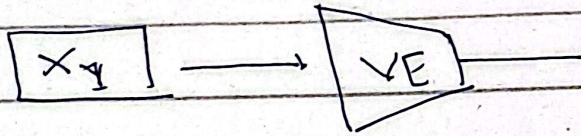


Lecture 6

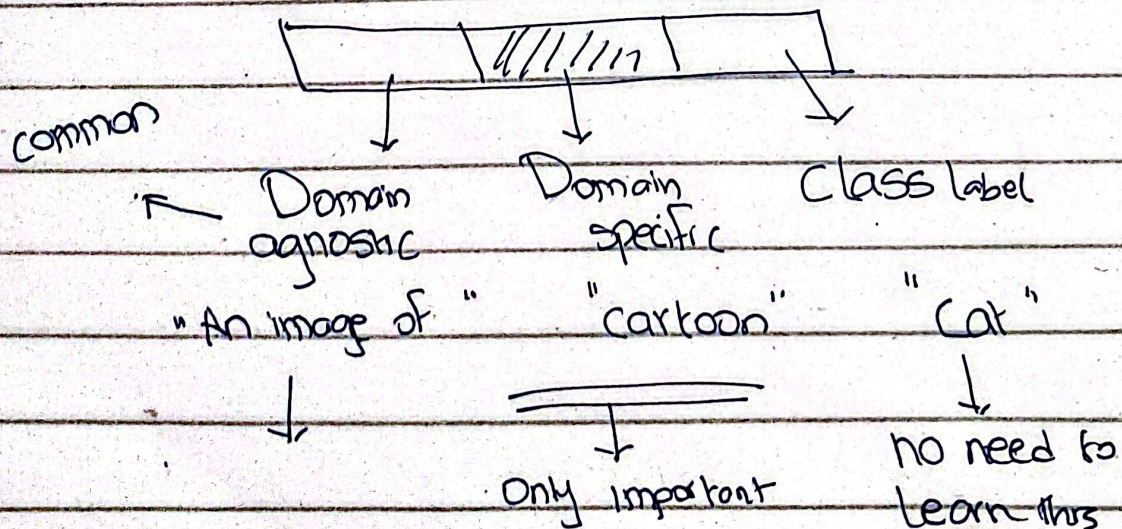
UDA - State of the art models

↳ Multimodal Models For UDA:



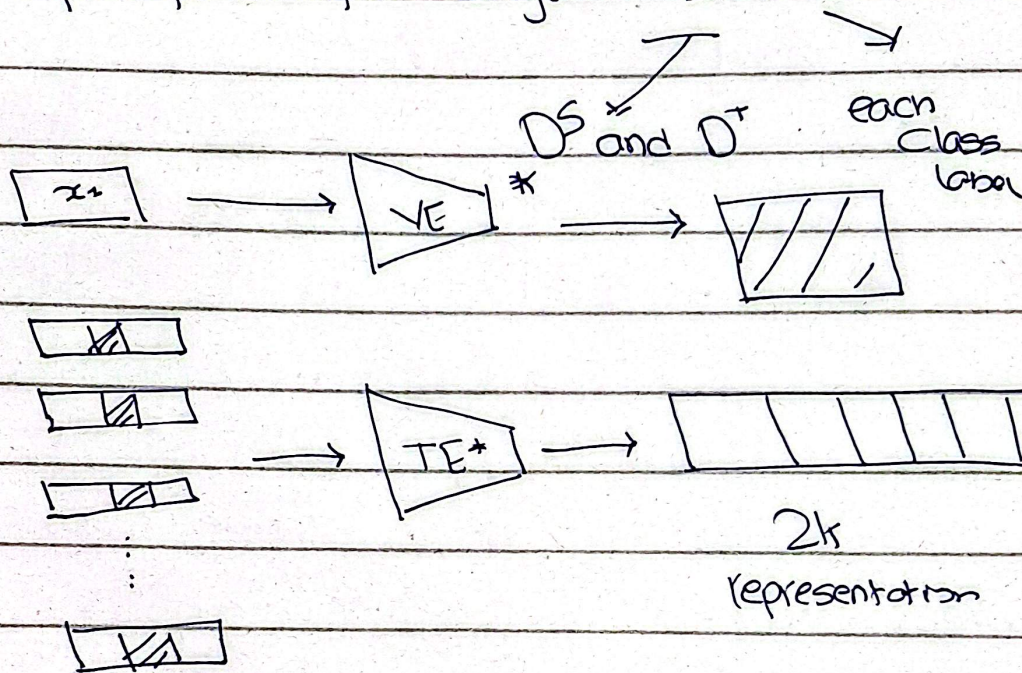
1. Prompt Learning:

Divide prompt to 3 parts



- Learn Domain specific by Freezing everything and only learn prompt through prompt examples

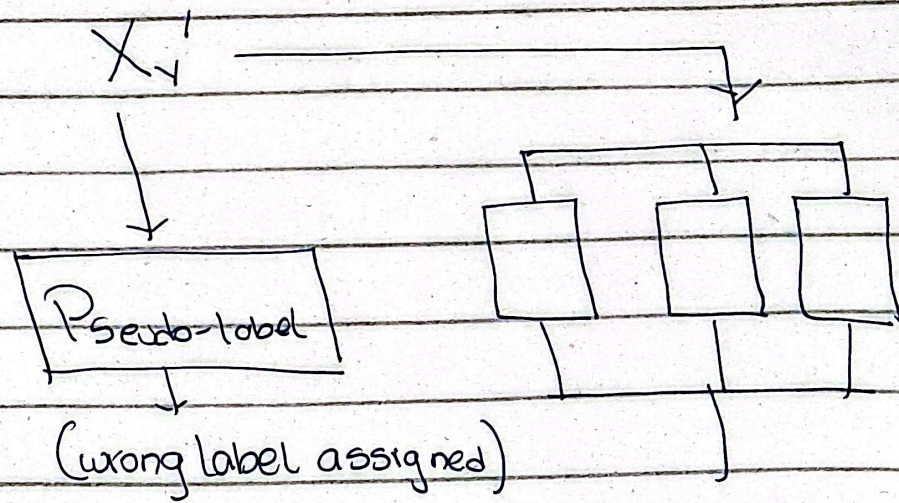
- prompt examples length = $2 \times k$



- Learning is done by taking similarity with each text representation and image representation

$$L_s = \frac{e^{\sum (x_v^s \cdot x_t^s)}}{\sum (//)}$$

Predicting Image Classification:



$$V_k^{t+1} \leftarrow V_k^t - \eta \left(\sum_{i=1}^S \nabla L_{S_i} + \nabla L_T \right)$$

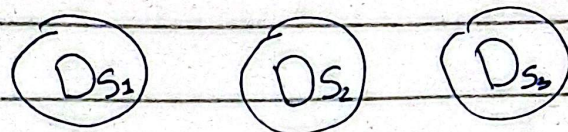
← ————— →
Gradient Conflicts

Gradient Conflict:

Whenever you have multiple gradient terms, and you are adding them to ~~to~~ in gradient descent, this usually leads to gradient conflict.

↓
CLIP solves it

* 1 Domain Generalisation (DG):



Same class label in each domain
↓
only covariate shift

$$\{x_i, y_i\}_{i=1}^n$$

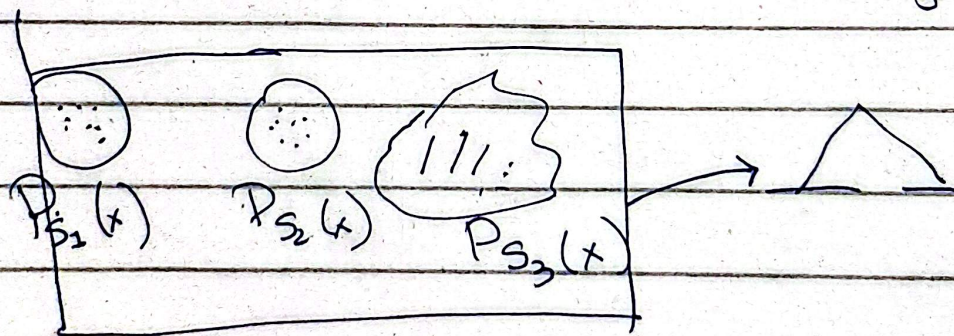


no data available

$$L_T(f_0) \leq \sum_{i=1}^S L_{S_i}(f_0) +$$

$$d_W(P_T(x), \Delta) + P + \text{constant}$$

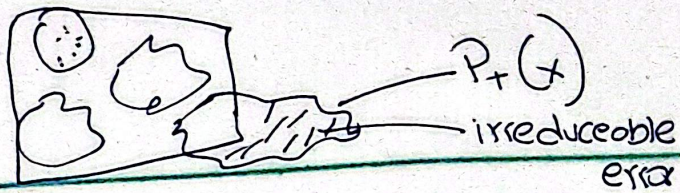
↓
ignore



Δ : bounding corners of each source domain

$d_W(P_T, \Delta)$: reducing distance here

Assumption:



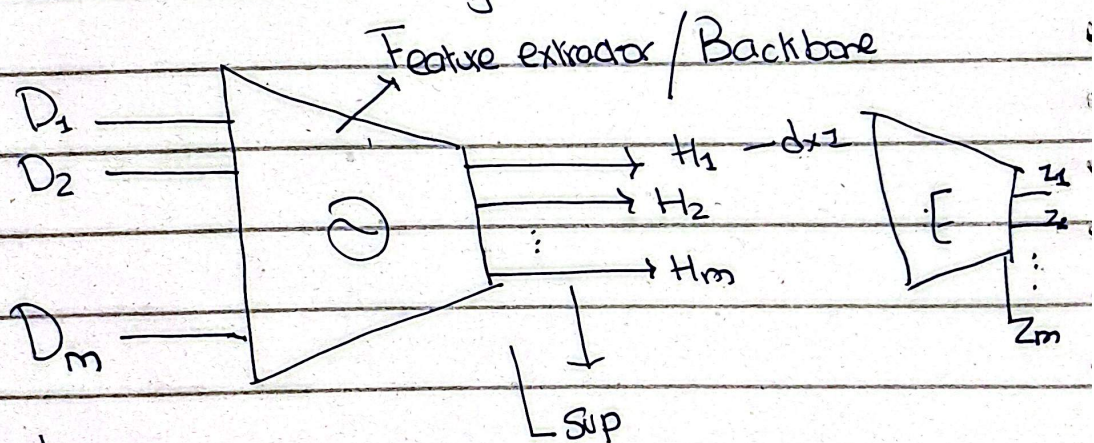
$P_+(x)$ will be correlated with Δ

Such that there might be some overlap between them

- Focus on causal features

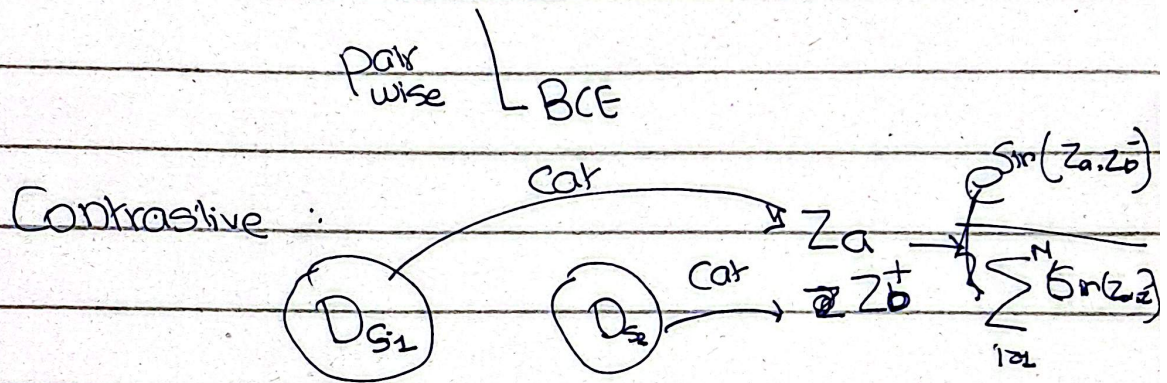
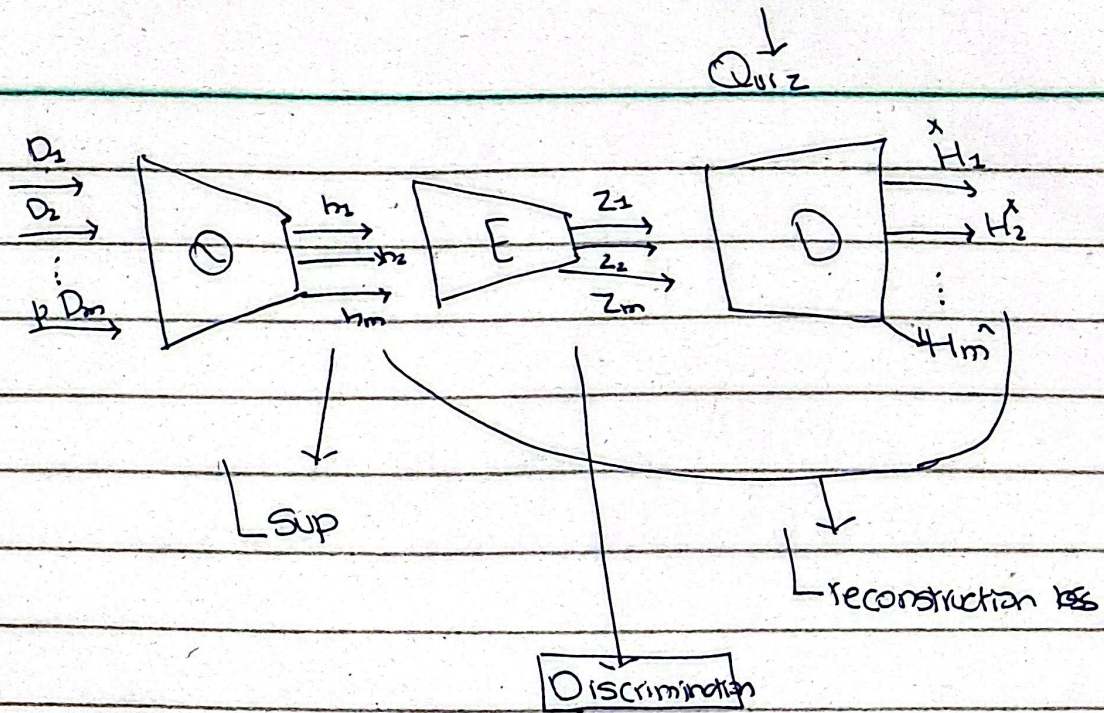
radius

→ The less the radius of Δ , meaning the more the compressed Δ is, the better the generalisation



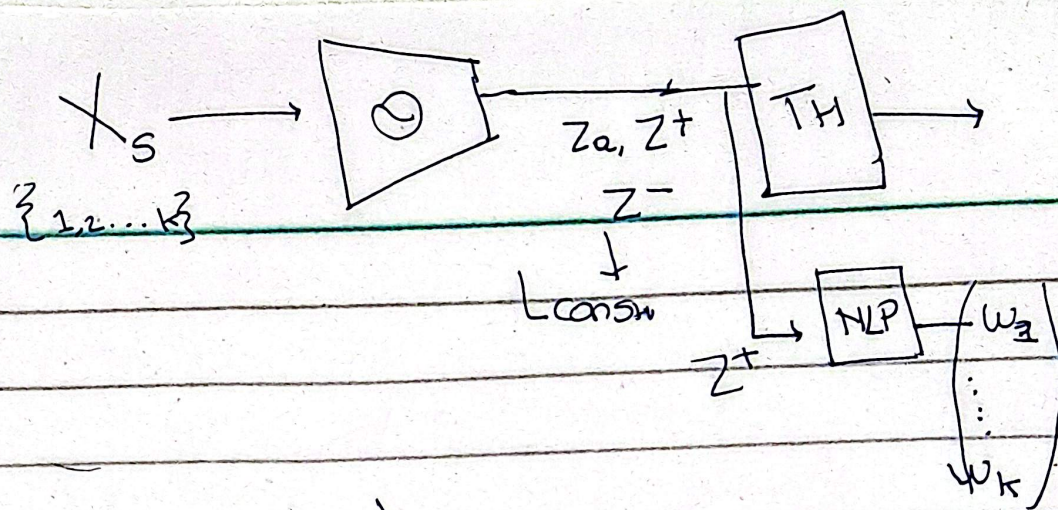
↓
Taking an image from each domain and pass and get $H_1, H_2 \dots H_m$, pass it through E to get latent features

Scenario — How to apply contrastive?



$z_a, z_b \rightarrow$ hard positives.

— Gradient push from these 2 from source domains is equal to gradient push coming from all negatives

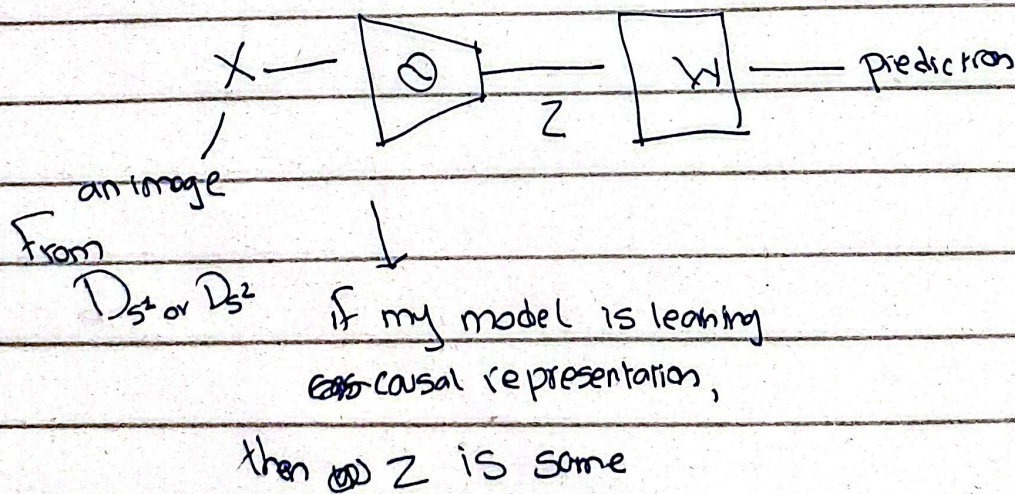


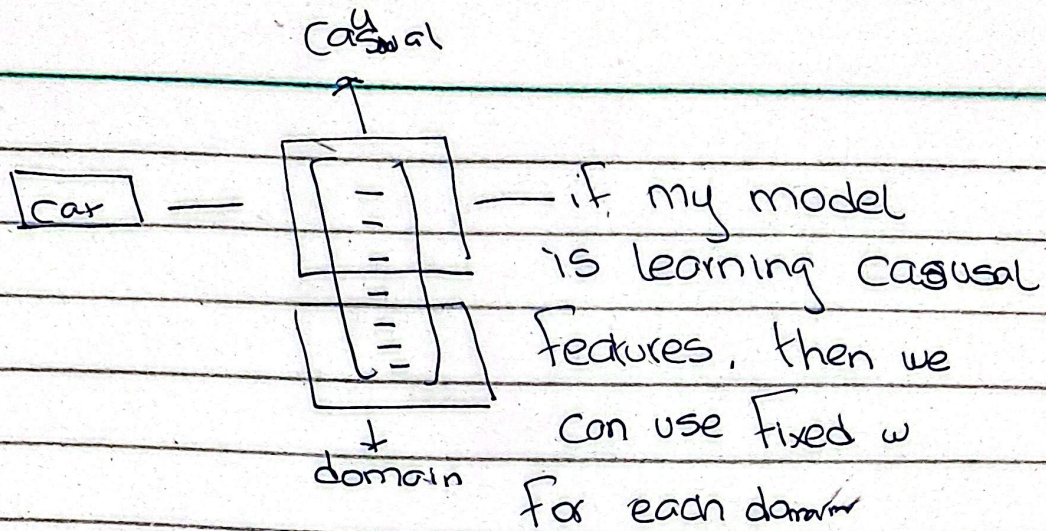
- These $\begin{pmatrix} w_1 \\ \vdots \\ w_k \end{pmatrix}$ proxies are fixed anchors for each ~~domain~~ class in latent space

$$\frac{e^{sm(Z_a, w_{cat})}}{\sum_{i=1}^N e^{sm(Z_a, Z^-)}} \quad \text{--- reduces hard positive and by learning average cat across all the domains}$$

- Techniques for DG:

1. IRM - invariant Risk minimization

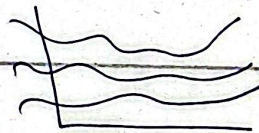




$$L_{IRM} = \sum_{i=1}^S L_{CE_i}(x_i, y_i) + \left\| \nabla L_{CE} \right\|_2$$

(new lecture)

W/ common minima for all domain loss



- basically

~~we want~~ ∇L_{CE} with respect to w^* is

0 and to make a vector

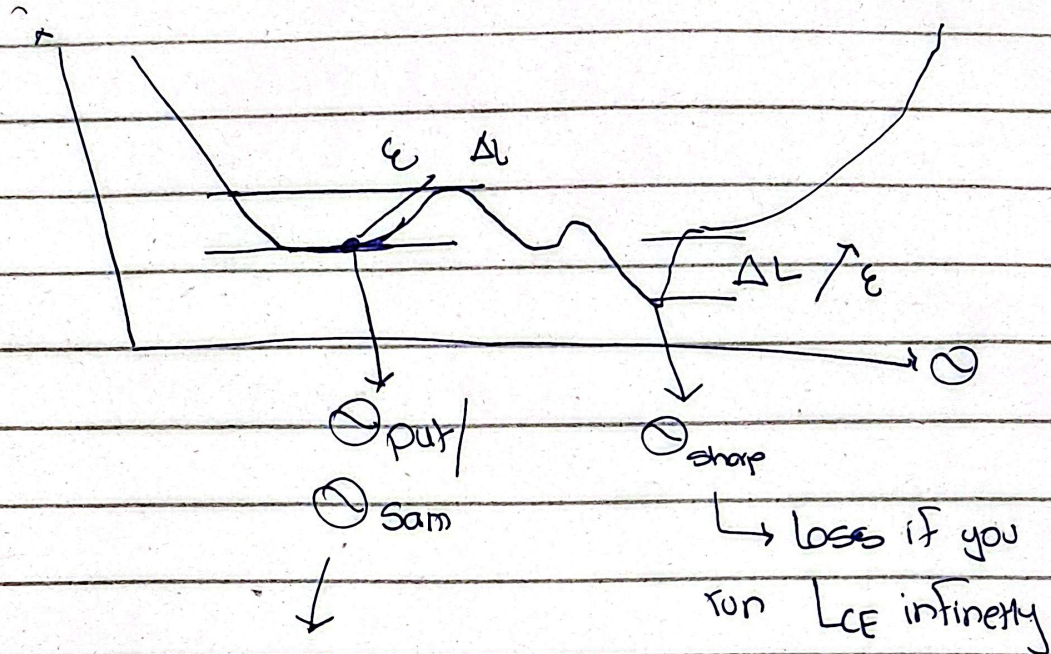
0 we take a gradient norm

never exist

Perfect ideal

LCE alternatives, CF doesn't ~~for~~ consider
inversions

2. Sharpness Aware Minimization: [SAM]



This point is more generalisable because CE is a greedy

→ Slight change in θ in

- θ_{sharp} will make loss

changed abruptly

- θ_{sam} is constant, so will
remain same

$$\min_{\Theta} \max_{\|\epsilon\| \leq \rho} L_{CE}(\Theta + \epsilon)$$

ΔL : perturbation

$$\downarrow$$

if $\Theta_s + \epsilon \quad \|\epsilon\| \leq \Delta L$

- basically we have ΔL (noise) and in Θ_s , we

see a

- basically we want to be in ΔL , like we want

to see how much noise ~~do~~ $[\epsilon]$, do we need

to add that noise is less than ΔL

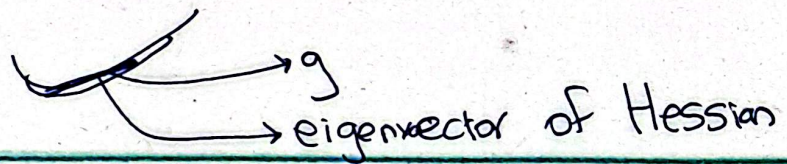
$\downarrow$

in Θ_s , — by adding a small amount of ϵ
we reach ΔL

in Θ_{nom} — by adding a large amount
of ϵ , we reach ΔL

↳ not adding idea, because if $\Theta = 1$ bill polymers
 $\epsilon = 1$ b polymers

main



$$E = \rho \frac{g_0}{\|g_0\|} +$$

